

# What a Small From-Scratch Financial Agent Can Learn from Feedback: Measured Wins, Failures and Traps

Rishabh Kapur

Department Name, Institution Name

City, Country

Email: kapur.rishabh13102003@gmail.com

**Abstract**—We ask whether an agent can improve its decisions from interaction, outcomes and feedback without continually retraining its underlying model. We built ALFA, a finance advisory agent, entirely in a NumPy automatic-differentiation library written for the project, with no external model and no API keys. It combines a 7.47M-parameter grounded text generator, a volatility head, a retriever and a tokenised generative model of daily returns. We report every measurement taken, including the failures. Feedback that moves only a threshold, with the model frozen, halves the gap between promised and delivered precision (16.7 to 8.5 points after 60 labelled answers). Preference fine-tuning on community votes did not help: plain DPO raised held-out pair agreement from 58.9% to 60.9% while the unsupported-figure rate of free generation rose from 59.4% to 92.9%. The cause was the data: 51.0% of 62,155 question-answer rows state a figure their own retrieved passages do not contain. Filtering those rows out with the server’s own guardrail cut answers containing an invented figure from 41.0% to 11.3% on held-out rows. On prices, direction is not forecastable (35.7% against a 36.0% majority class), but the distribution is. An 876k-parameter return generator beats GARCH(1,1)-t on the one-step log score by  $0.0323 \pm 0.0032$  nats per return over 80,576 held-out returns. Its sampled paths reproduce the tail index and the leverage effect. An online temperature that learns from each realised return, weights frozen, improves the log score by a further  $0.0078 \pm 0.0017$  nats. We also catalogue the measurement traps that produced false conclusions along the way.

**Index Terms**—Learning from feedback, grounded generation, preference optimisation, faithfulness, volatility forecasting, GARCH, generative models, proper scoring rules, abstention.

## I. INTRODUCTION

Most published stock-prediction systems report accuracy on a single target, often with little control for leakage [1]–[3]. Most feedback-learning systems retrain a large pretrained model [4]–[6]. We studied a harder and smaller setting. The agent is built from scratch on one CPU. It may not call an external language model. It must answer conversational finance questions without inventing figures. Its research question is whether it can get better from interaction, outcomes and feedback *without* continually retraining its underlying model.

The system separates four concerns. *Knowledge* is a frozen encoder plus retrieved documents. *Reasoning* is code. *Policy* is the only part that learns. *Memory* is per user. Policy learning can therefore live in a threshold, a calibrator, a sampling temperature or a reward, as well as in weights. We measured each.

Our contributions are:

- 1) A complete from-scratch system and its evaluation discipline: date-based splits, one margin rule for every promotion, generate-and-check before loss, and point-in-time serving pinned by tests.
- 2) Evidence that feedback moving only a policy parameter measurably improves the agent, with the model frozen (Sections V and VIII-D).
- 3) A negative result on preference learning, traced to its cause: an unfaithful corpus, which a rule-based verifier measures directly (Section VI).
- 4) A tokenised generative return model that beats GARCH-t on the log score and reproduces stylized facts GARCH cannot, and a precise account of where it does not win (Section VIII).
- 5) A catalogue of the measurement traps we hit, each of which produced a false conclusion before it was caught (Section IX).

## II. RELATED WORK

**Deep learning for stock prediction.** CNN-BiLSTM index predictors [1], reinforcement learning on chart images [2] and time-series-to-image encoders [3] report directional gains. A recent IEEE review [7] surveys the architectures. Large 2026 benchmarks are sceptical. Time-series foundation models forecasting hourly returns for 1,000 US equities collapse to near-flat predictions [8]. Across 43 models, generic accuracy “does not necessarily translate into useful financial forecasts” [9].

**Volatility and distributions.** GARCH [10] with Student-t innovations [11] remains the reference. Neural alternatives are older than deep learning [12] and continue in hybrid forms [13]. Chronos [14] showed that a plain language model learns time series once values are scaled and quantised into tokens.

**Synthetic financial series.** GANs generate synthetic series in smart grids [15] and finance [16]. Quant GANs [17] evaluate against the stylized facts of Cont [18] and report training that does not converge. We use their metrics and a likelihood objective that does converge.

**Learning from feedback.** Interactive reinforcement learning from human feedback [4] and RLHF for code [5] learn from judgements, and RLHF has known vulnerabilities [19].

DPO [6] removes the reward model. Adding the chosen answer’s likelihood [20] anchors it. DeepSeekMath [21] writes SFT, DPO, PPO and GRPO as one gradient that differs only in data source, reward and weighting. Verifiable rewards [22], reward-ranked fine-tuning (RAFT) [23], self-taught filtering [24] and dynamic sampling [25] replace the learned judge with a check. Continual fine-tuning forgets [26], which motivates learning outside the weights.

**Faithful retrieval-augmented generation.** RAG [27] grounds answers in retrieved text, and BM25 [28] remains a strong retriever. Rewarding correct answers does not buy faithful citations [29]. A faithfulness-only reward is won by terse replies, so financial RAG needs an informativeness term [30].

**Scoring.** Strictly proper scoring rules [31] reward only honest distributions. Prequential evaluation [32] scores each forecast before its outcome is learned from. Paired comparisons follow Diebold and Mariano [33]. Tail indices use Hill’s estimator [34].

### III. SYSTEM AND DATA

#### A. Constraints

The constraints are fixed. There is no external model and no API key. Deep-learning frameworks are banned by an import test, so every model is built on the project’s own NumPy code: 445 lines of autograd and 928 with the layers and optimiser, checked against finite differences. There is no look-ahead: every split is by date, and every served figure is rebuilt from bars up to its as-of date, pinned by tests that mutate the data after that date. No real money is traded. Everything runs on one laptop CPU.

#### B. Data

The collector fetches only public bulk endpoints, official APIs and RSS feeds. It honours `robots.txt`, allows one request at a time per host, and records the URL and licence of each file in a manifest of 177,358 entries (8.1 GB). The collection comprises:

- **Prices:** daily OHLCV for 101 US and Indian equities. Histories start between 1962 and 2017 and end on 18 September 2026.
- **Question-answer pairs:** Stack Exchange (*money*, *economics*, *quant*), giving 62,155 supervised pairs and 24,639 vote-ranked preference triples.
- **Pretraining text:** 798,784 sequences of 128 tokens, about 102M tokens, under an 8,000-piece WordPiece vocabulary.
- **News:** 1,481 deduplicated articles from 55 feeds.

#### C. Models

Fig. 1 shows the agent. A 4-layer, 256-wide encoder was pretrained by masked language modelling. The run took 49,232 steps and 11.09 h, reaching a validation perplexity of 20.6. The grounded generator (7,472,192 parameters) adds a 2-layer decoder that cross-attends to four retrieved 128-token passages, fused in the decoder. A guardrail extracts every

figure in an answer and refuses the turn if any figure is absent from the evidence. The price head is a 202,755-parameter GRU over 128-bar windows. The return generator is described in Section VIII.

#### D. Evaluation discipline

Three rules apply throughout:

- 1) **One margin rule.** A candidate replaces what is served only if it beats it by more than one standard error of the difference, measured on identical rows.
- 2) **Generate, then check.** A text model is judged by answers it writes, not by loss.
- 3) **Held-out means untouched.** Every quoted figure comes from a period, phrasing or split that chose nothing.

The S14 faithfulness standard requires unsupported figures below 2%.

### IV. GROUNDED GENERATION

#### A. Copying is a late phase transition

On a synthetic advisory set of 225,309 rows, generated from computed indicators, training loss sat at 0.28 through step 3,000 and then collapsed to 0.014 by step 5,400. At step 3,000, generate-and-check found 0% exact match and an unsupported figure in every answer. At step 4,000 it found 45.8% exact match and 10.5% unsupported figures. The 0.28 plateau is the entropy of each field’s marginal prior: the model emits the distribution of *close*, not this row’s *close*. It is not a capacity floor.

#### B. The decoder reads its evidence

On the converged model, hiding the evidence costs +1.042 nats and swapping in another row’s evidence costs +1.946. The swap costing more than the blank shows that the model reads what it is given. On the real Stack Exchange task, by contrast, hiding every passage costs only +0.018. Only 30.7% of an answer’s content words appear anywhere in its input, so ignoring the evidence is the loss-minimising strategy there.

#### C. Sentence shape, not vocabulary

Unseen phrasings stalled at a loss of 0.70 while trained phrasings reached 0.011. A probe over 190 composed phrasings located the cause. The pretrained encoder places an unseen *word* correctly in 100% of cases, but an unseen *frame* in only 68%. Task fine-tuning halves the latter to 32%, which is forgetting [26] measured directly. Two fixes followed. Rebuilding the data from 46 to 190 phrasings cut the unseen floor to 0.2821. Freezing the encoder then cut it to 0.1717, and unsupported figures on unseen phrasings from 4.7% to 0.7%. The promoted checkpoint states 4 of 771 figures (0.5%) unsupported on reworded questions, with 36.0% exact match.

#### D. Decoding

Greedy decoding repeated 73% of its 4-grams, against 1.7% for human answers. Nucleus sampling at temperature 0.9 and top-*p* 0.9 measures 1.6% repeats and 0.4% invented words.

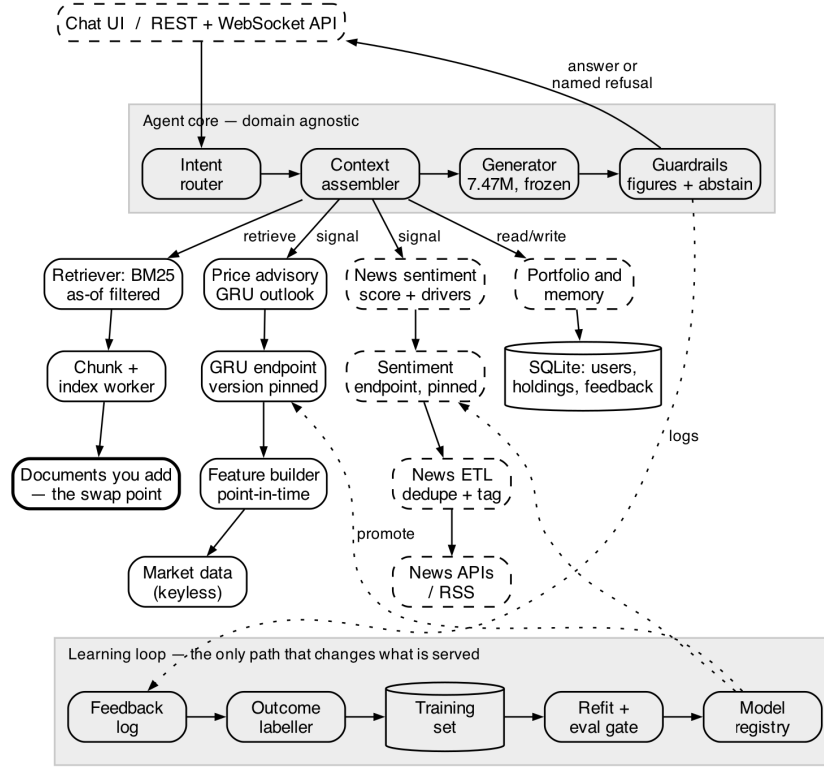


Fig. 1. ALFA. The agent core (top) routes a question, assembles evidence, generates with frozen weights and screens the answer. The learning loop (bottom) is the only path that changes what is served, and every change passes an evaluation gate.

## V. LEARNING WITH THE MODEL FROZEN

### A. Calibration and abstention

The raw confidence claims 0.99 on answers that are right 31% of the time. A two-parameter Platt fit lowers the calibration error from 0.683 to 0.121 over 25 splits. Its ranking AUC is  $0.740 \pm 0.039$ , which caps any threshold or reranker built on it. The agent is then given a precision promise (“answer only when at least  $p$  likely to be right”) and solves for the confidence cut from the feedback it has.

Fig. 2 scores the *miss*: the gap between promised and delivered precision, on rows the cut was not fitted on. At a 60% bar, 60 feedback rows cut the miss from 16.7 points (the hand-picked cut) to 8.5. Answering everything misses by 29.2. At an 80% bar under the agent’s labels, the miss falls monotonically from 13.3 to 6.3 and is still falling at 118 rows. No weight changed.

In a 20-turn session over a live WebSocket, the fitted cut moved from 0.998 to 0.971 at the tenth judged turn. It *loosened*, because the log did not justify withholding anything. The agent spoke on 14 of 17 judged turns and was right on 13 of them (92.9%).

### B. Routing with frozen weights

Summing the frozen encoder’s cosine with the share of the question’s term weight that known phrasings cover raises routing accuracy on an unseen sentence shape from 67% to

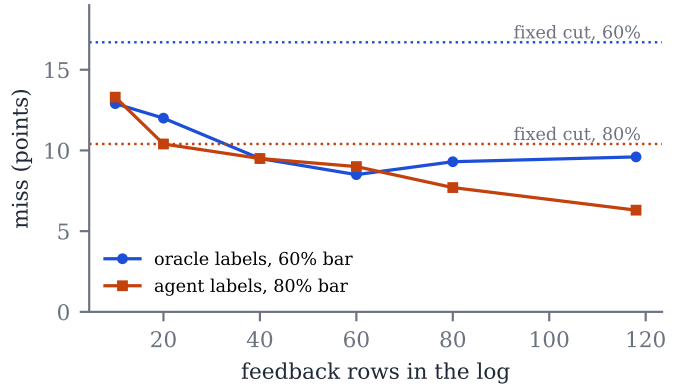


Fig. 2. Feedback moves only the abstention threshold, and the delivered precision approaches the promised one. Dotted lines: the hand-picked cut with no feedback.

90%. Rewriting a question into its routed trained phrasing raises held-out exact match from 33.5% to 74.5%. Twelve of twelve out-of-domain questions are refused. No parameter was fitted.

### C. Best-of- $k$ does not pay

Eight samples from the advisory generator contain 1.77 distinct answers. Best-of-8 by confidence scores 27.8% and by grounding tier 28.5%, against 28.0% for a single sample

TABLE I  
PREFERENCE TRAINING ON VOTE LABELS (HELD OUT)

|                                 | Frozen | DPO   | DPO + NLL |
|---------------------------------|--------|-------|-----------|
| Pair agreement (2,000)          | 58.9%  | 60.9% | 59.2%     |
| Repeated 4-grams                | 1.7%   | 30.8% | 0.3%      |
| Unsupported figures             | 59.4%  | 92.9% | 74.3%     |
| Answers with an invented figure | 39.0%  | 59.0% | 40.0%     |

on 600 rows. A model whose token distribution is nearly a delta cannot be improved by sampling it more.

## VI. PREFERENCE LEARNING, AND WHY IT FAILED

### A. How good are the labels?

A blind reader judged 60 held-out vote-ranked pairs with the sides shuffled, and agreed with the votes on 40 (66.7%, roughly  $\pm 6$  points). A third of the training labels are judgements a careful reader would not make. The reader was the AI assistant used in development, not a domain expert.

### B. Plain DPO collapses

One epoch of DPO ( $\beta = 0.1$ , length-normalised, reference fixed) on 22,379 pairs lowered the held-out loss from 0.6931 to 0.6617. Agreement rose from 58.9% to 60.9%, but McNemar’s test on the flips (419 right, 379 wrong) gives  $p = 0.17$ . Free generation collapsed (Table I): every answer opened with about thirty copies of “you”. The mechanism is measurable. Among 5,226 frequent tokens, “you” has the 4th-highest over-representation in preferred answers (1.929% against 1.867%). Raising one frequent unigram raises the mean log-probability of the chosen side slightly more than the rejected side on nearly every pair, which wins the objective without reading either answer. Loss and agreement improved throughout, and neither could see the failure.

Adding the chosen answer’s negative log-likelihood [20] fixed the collapse (0.3% repeats) but learned nothing. At the default weight of 1.0 the supervised gradient is about 20 times the ranking gradient, whose magnitude is bounded by  $\beta$ . Neither run beat the frozen checkpoint, and nothing was promoted.

### C. The root cause is the corpus

Among 4,000 sampled supervised rows, 2,390 state at least one figure, and only 116 of those (4.9%) have every figure in their retrieved passages. On the full build, 31,719 of 62,155 rows (51.0%) state a figure their passages do not contain. The generator’s unsupported-figure rate is this data learned correctly. Both sides of a preference pair are drawn from the same corpus, so no ordering between them expresses grounding.

### D. From labels to a check

Following the verifiable-reward literature [22], [23], the reward is the guardrail itself. An answer that states an unsupported figure scores 0. A refusal scores 0.1. A grounded answer scores  $1 + 0.5 \times$  its number of grounded figures, so

TABLE II  
GENERATOR RETRAINED ON THE FILTERED CORPUS (300 CLEAN HELD-OUT ROWS)

|                                 | Original    | Filtered corpus |
|---------------------------------|-------------|-----------------|
| Figures stated                  | 474         | 132             |
| Unsupported figures             | 422 (89.0%) | 71 (53.8%)      |
| Supported figures               | 52          | 61              |
| Answers with an invented figure | 41.0%       | 11.3%           |
| Repeated 4-grams                | 0.9%        | 0.7%            |

going quiet cannot win, as RLKV warns [30]. Groups in which every sample scores the same are dropped, following DAPO [25]. A dry run on 96 questions with 8 samples each found 87 teachable groups, 9 with identical scores, none in which every sample invented a figure, and 8.00 distinct answers out of 8. The loop therefore has signal on that model. Run from the grounded generator described next, it did not pay. The grounded model states few figures, so 129–132 of 192 groups a round scored alike and only about 8 training steps a round remained. The first round read 65.6% to 54.9% unsupported on its 200 selection answers. On 300 independent rows it scored 63.0% against the frozen model’s 53.8%, with 44 supported figures against 61. That was selection noise, and the two later rounds were refused by a guard against winning the rate by saying less.

### E. Filtering the corpus with the same check

The same rule, applied to the training targets, keeps 30,436 of the 62,155 rows. A generator retrained on them was compared with the original on 300 held-out rows that neither saw in training. The two splits were drawn independently, and 949 of the original 2,000 validation rows had landed in the filtered training set, so those were removed first. The remaining rows lean towards questions whose passages lack the figures, which is why the original model scores worse on them than on the full split. Table II gives the result. Unsupported figures fell from 89.0% to 53.8%, and answers containing an invented figure from 41.0% to 11.3%. The drop is not the model going quiet: it states more supported figures than before, 61 against 52. The prose is still incoherent in both models, so faithfulness improved and fluency did not.

## VII. PRICES: DIRECTION, VOLATILITY, RETRIEVAL, NEWS

Table III summarises the results. Five-day **direction** is not learnable: the transformer scored 35.7% and the GRU 33.5%, against 36.0% for the majority class, with the loss flat at  $\ln 3$ . Five-day **volatility** is learnable. On all 28,676 held-out windows after 2021-01-01, the GRU scores 49.8% against 47.8% for bucketed trailing volatility, 6.8 standard errors. A 1.6M-parameter transformer tower scores 46.3%, below the one-line baseline, and averaging towers scores 48.9%. Serving uses the GRU.

**News** polarity agrees with the next day’s return  $49.6\% \pm 2.3\%$  of the time on 456 pairs, against a 52.6% base rate, so it is not wired into advice. On the publication day, however, positive minus negative return is  $+0.90\%$  at 2.6 standard errors. The lexicon reads headlines; only the forecast fails.

TABLE III  
PRICE AND TEXT SIGNALS, HELD OUT

| Task                         | Model | Baseline          |
|------------------------------|-------|-------------------|
| 5-day direction (3 classes)  | 35.7% | 36.0% majority    |
| 5-day volatility (3 classes) | 49.8% | 47.8% persistence |
| News sign vs next-day return | 49.6% | 52.6% base rate   |
| Recall@5, vector retrieval   | 1.5%  | 15.5% BM25        |

**Retrieval.** Without a contrastive objective, mean-pooled embeddings are anisotropic (mean norm 0.942 of 1), so vector recall@5 is 1.5% against 15.5% for BM25 [28]. The vector half was dropped. Over 519,135 chunks and 62,192 evaluable threads, BM25 reaches 22.9% recall@1, 34.2% at 5, 45.9% at 20 and 52.9% at 50. When the reference path quotes the retrieved passage rather than paraphrasing it, 200 real questions produced no invented figure. A graded paraphrase of the same passages needed the guard to catch 186 figures in 60 answers.

## VIII. A GENERATIVE MODEL OF RETURNS

### A. Model

We follow Chronos [14], with two changes forced by returns. First, returns are scaled by  $s$ , the mean absolute return over the 250 bars *before* the window. A scale taken over the window would divide each input by a number containing the returns being predicted. Second, the 256 bins are uniform in  $\text{asinh}(r/s)$  over  $\pm 30$  units. This is a mu-law compander: 0.03 units wide at the centre, 0.16 at five units, and reaching a 39% day at the edge (the median unit is 1.30%). Only 88 of 842,624 returns clip; the largest, at 410 units, is a data error.

A 4-layer causal transformer (876k parameters, context 128) is trained with cross-entropy, which is maximum likelihood and the log score, a strictly proper rule [31]. Spreading each bin’s probability uniformly over its width gives an exact density,

$$f(r) = \frac{p_k(r)}{s (\sinh e_{k+1} - \sinh e_k)}, \quad (1)$$

so the model’s score is directly comparable with a Gaussian’s or GARCH’s on the same returns. Paths are sampled at temperature 1, because a nucleus cut would delete the tail. The model was trained on 691,200 windows before 2021, selected on 2021–22, and tested from January 2023 to September 2026.

### B. Baselines

GARCH(1,1)-t [10], [11] is fitted by variance targeting, with  $\omega = \bar{v}(1 - \alpha - \beta)$ , and a grid over  $(\alpha, \beta, \nu)$ . It lands on the textbook values  $\alpha = 0.08$ ,  $\beta = 0.90$ ,  $\nu = 4$ . The second baseline is a Gaussian at the trailing 20-day volatility. Paired differences use each window as one block [33], because returns inside a window share a regime.

### C. Results

Table IV gives the results. The generator wins the one-step log score by 10 standard errors. It ties both baselines on 5- and 20-day path CRPS, estimated in energy form as

TABLE IV  
RETURN GENERATOR, TEST PERIOD 2023–SEP 2026

| Measure                       | Model                  | GARCH-t | Real   |
|-------------------------------|------------------------|---------|--------|
| – log score gap (nats/return) | –0.0323 ± 0.0032       |         |        |
| 5-day CRPS gap                | –0.0007 ± 0.0006 (tie) |         |        |
| Hill tail index               | 2.98                   | 2.35    | 3.08   |
| EMD, 1-day returns            | 0.032                  | 0.414   | 0      |
| EMD, 5-day returns            | 0.145                  | 1.049   | 0      |
| Leverage, lags 1–5            | –0.025                 | –0.002  | –0.017 |
| acf of $ r $ , lags 1–5       | 0.076                  | 0.092   | 0.047  |
| Direction hit rate            | 51.0%                  | –       | 51.2%* |

\*Always calling the more common side. 80,576 returns scored.

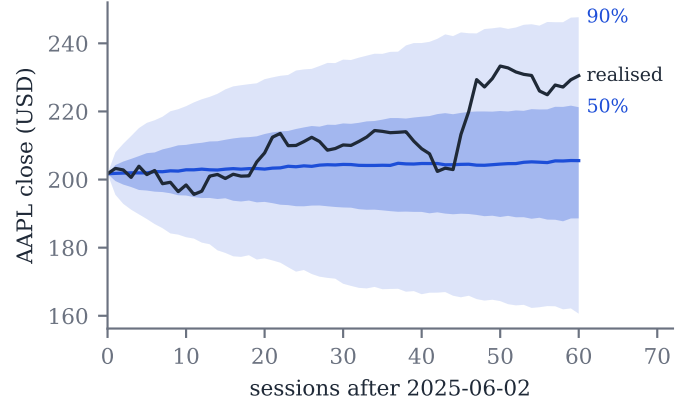


Fig. 3. 2,000 sampled AAPL paths from 2 June 2025, inside the test period (shaded: central 50% and 90%), and the path that followed. The fan states width, not direction.

$\mathbb{E}|X - y| - \frac{1}{2}\mathbb{E}|X - X'|$  over 50 paths at 150 dates. Its paths match the real tail index and the distribution of daily and weekly returns far better than GARCH. They reproduce the leverage effect, which symmetric GARCH cannot. Volatility clustering is present, and the model’s predicted spread correlates 0.864 with the context’s recent volatility, but it is too strong. Direction from  $P(r > 0) > 0.5$  is right 51.0% of the time against 51.2% for always calling the more common side. Fig. 3 shows a fan drawn from a date inside the test period, with the path that followed.

### D. Learning from outcomes, weights frozen

A temperature  $T$  rescales the logits  $z$ . After each realised return it takes one exact gradient step on the log score,

$$\frac{\partial (-\log p_y)}{\partial \log T} = \frac{z_y - \mathbb{E}_p[z]}{T}. \quad (2)$$

Replaying the test period day by day, each forecast is scored before its outcome is learned [32]. Over 93,437 returns this improves the log score by  $0.0078 \pm 0.0017$  nats per return, 4.6 standard errors blocked by month. The frozen model was already calibrated: its central 90% interval held 89.9% of outcomes.  $T$  ranged from 0.51 to 2.88. This is the research question answered on the numeric side: the forecast improves from feedback while the model under it stays fixed.

### E. Serving

The server draws paths with the generator only while its own artifact records a win over GARCH of more than one standard error. Otherwise GARCH draws them. Every response carries a sentence stating that the paths show width, not direction. Synthetic data is written with a sidecar recording the drawer, the checkpoint digest, the seed and the as-of date.

### IX. CHALLENGES AND MEASUREMENT TRAPS

Each item below produced a wrong conclusion before it was caught.

- 1) **Two numbers, two row sets.** The price head once “tied” persistence, 46.9% against 47.0%. The head had been scored on the first 1,920 windows, which covered 7 of 101 tickers.
- 2) **A probe that leaked.** An intent probe scored 90% on an encoder with no task training, because a reserved phrasing matched another reserved phrasing.
- 3) **Loss picks worse models.** A loss-selected checkpoint generated with 17.2% unsupported figures, against 2.4% for the final one, and 77.0% exact match against 94.5%. All promotion now rests on generation.
- 4) **Plateaus that look like limits.** Judged at step 3,000, the grounded generator “could not copy”. It could by step 4,000.
- 5) **Noise minima.** Scored on 64 generated answers, one DPO run promoted a checkpoint at 79.5% unsupported figures in the sequence 93.3, 90.5, 79.5, 97.9, 100.0.
- 6) **The machine slept.** Pretraining launched without a sleep inhibitor logged 15.5 h of wall-clock time but only 39 min of CPU, reaching step 1,400 of 49,232.
- 7) **Serving bugs invisible to tests.** The guard refused grounded answers because decoding put a space between a minus sign and its digits, reading  $-5.6\%$  as  $5.6\%$ . Every existing test wrote the sign attached, so the suite was green. Separately, token ids stored as uint16 wrap the  $-100$  ignore index to 65,436, a valid vocabulary index, unless widened first.
- 8) **Reward hacking.** DPO won its objective by writing “you” (Section VI).
- 9) **Dirty data.** Price files contain returns of 410 scale units and years of zero movement. Both are rejected loudly rather than repaired.
- 10) **Compute.** Everything ran on CPU. The generator trains at about 1.3 steps/s, and one preference run takes about an hour, so every experiment was budgeted before launch.

### X. THREATS TO VALIDITY

The 101 tickers were chosen in 2026, so the price history carries survivorship bias. Nothing here estimates returns net of transaction costs, and no trading claim is made. Most runs use one seed. The 60-pair label audit and the agent-side feedback labels were written by an AI assistant, not a domain expert. The feedback logs are small: 118 rows for the abstention curve and 20 turns for the live session. The return test period (2023

to 2026) is one market regime. Online temperature learning was run at a single step size.

### XI. CONCLUSION

A from-scratch agent can learn from feedback without retraining its model. In this system that learning showed up in three places: a threshold solved from outcomes, a router reading frozen weights, and a temperature moved by each realised return. Each gain is measured and beats a stated baseline. Changing the weights from community preferences did not help, and the reason was the data, not the algorithm: half the corpus states figures its evidence does not contain, and removing those rows cut answers with an invented figure from 41.0% to 11.3%. On prices, the distribution is learnable and the direction is not. A small tokenised generator beats GARCH-t on the log score and generates paths with realistic tails and leverage, while its multi-day forecasts only tie. The most transferable lesson is methodological. Every false conclusion in this project came from measurement, not modelling, and each was caught by insisting on identical rows, held-out periods and generate-and-check.

### ACKNOWLEDGMENT

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